

Microsoft Azure Databricks

1. What is Azure Databricks?

Azure Databricks is a cloud-based big data analytics platform built on top of Apache Spark, jointly developed by Microsoft and Databricks. It is designed for data engineers, data scientists, and analysts to process large amounts of data, build machine learning models, and run analytics — all in one collaborative environment hosted on Azure.

Think of it as a powerful notebook-style workspace where you write Python, SQL, Scala, or R code to analyze massive datasets.

2. What are the Advantages of Using Azure Databricks?

- **Speed** — Built on Apache Spark, it processes data much faster than traditional tools
 - **Collaboration** — Multiple team members (data engineers + data scientists) can work together in shared notebooks
 - **Scalability** — Automatically scales compute resources up or down based on workload
 - **Azure Integration** — Works seamlessly with Azure Data Lake, Azure Blob Storage, Azure Synapse, and Power BI
 - **Unified Platform** — Handles data engineering, machine learning, and analytics in one place
 - **Supports Multiple Languages** — Python, SQL, Scala, R all supported
 - **Auto-managed Clusters** — No need to manually set up or manage Spark clusters
 - **Security** — Enterprise-grade security with Azure Active Directory integration
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3. What is a Workspace?

A Workspace in Azure Databricks is your personal or team environment where all your work lives. It is like a project folder that contains:

- **Notebooks** — where you write and run your code
- **Clusters** — compute resources that run your notebooks
- **Libraries** — Python packages or JARs you install

- **Jobs** — scheduled automated pipelines
- **Data** — connections to your data sources

Every user or team gets a workspace, and you can organize everything inside it into folders, share notebooks with teammates, and manage access permissions.

4. What is EDA — Exploratory Data Analysis?

EDA is the process of examining and understanding a dataset before building any models or drawing conclusions. It is the first step any data analyst or data scientist does after receiving raw data.

What you do in EDA:

- Check the shape of data — how many rows and columns
- Look for **missing values** and decide how to handle them
- Find **duplicates** and remove them
- Understand data types — numeric, categorical, date, etc.
- Look at **distributions** — min, max, mean, median, standard deviation
- Find **outliers** — unusual values that may skew results
- Check **correlations** — which columns are related to each other
- Visualize data using charts — bar charts, histograms, box plots, scatter plots

In **Azure Databricks**, EDA is done inside notebooks using Python libraries like **Pandas**, **Matplotlib**, **Seaborn**, and **Plotly** on top of Spark DataFrames.
